Dynamic programming (class 9)

1. (a) Let M be a partial matching of indices for sequences $X \in \Sigma^*$ and $Y \in \Sigma^*$ of lengths n and m. M is called a *join*, if $j < j'$ holds for any $(i,j),(i',j') \in M$ and $i < i'$.

    Example: $X = gatcggcat$ and $Y = caatgtgaatc$.

g	-	a	t	c	g	-	g	c	a	t	-
c	a	a	t	-	g	t	g	a	a	t	c

    The cost of a join: $g > 0$ for any unmatched letter plus the sum of the rewriting costs $c(x,y)$ $(x, y \in \Sigma)$.

    Find a $O(nm)$ DP algorithm for determining the minimum cost.

    (b) Run the algorithm for *bab* and *abbaa* if $g = 2$ és $c(a,b) = 3$.

2. Examples for DP algorithms in BSc studies:

    (a) Floyd-Warshall algorithm (shortest paths for all pairs of nodes).

    (b) CYK algorithm (membership problem of CF grammars).